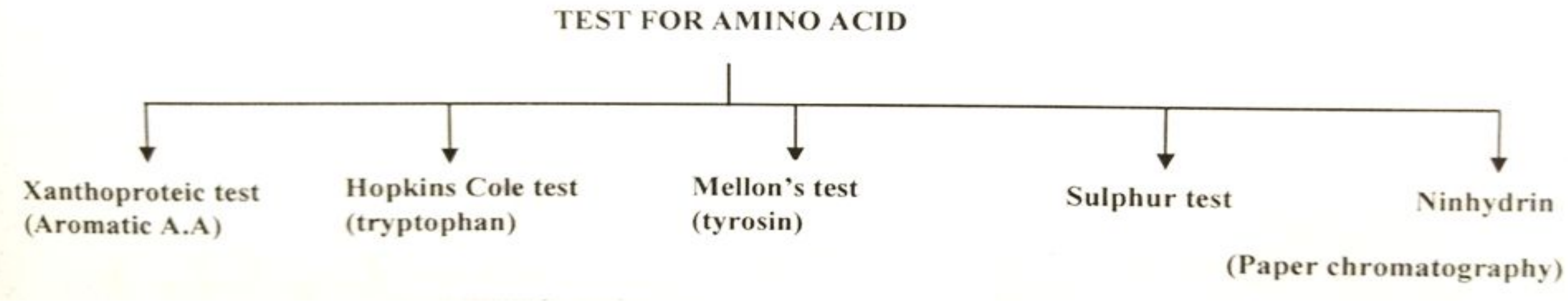
DR. HUMAIRA AMAN ALI

General test for detection of Proteins

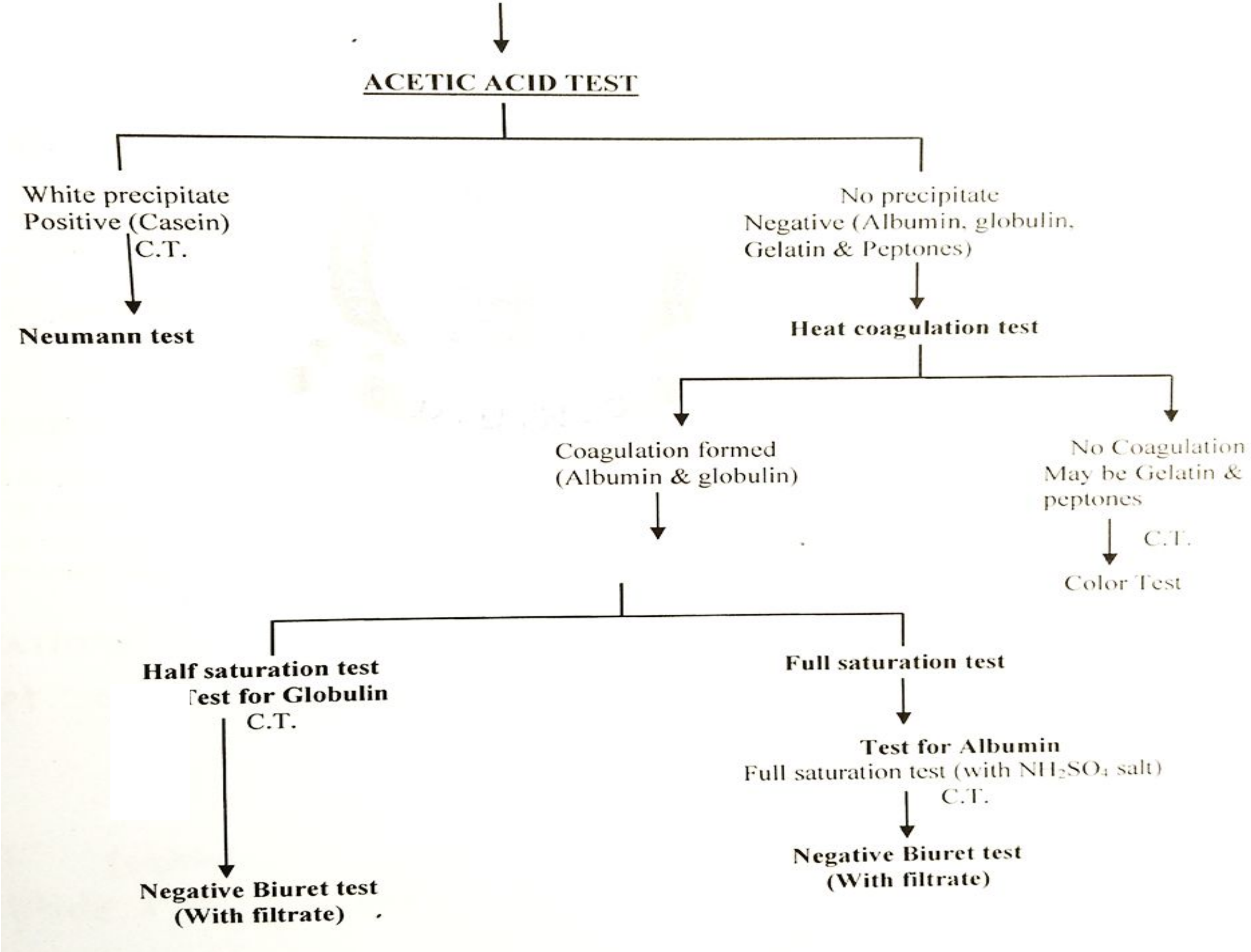
Biuret test (peptide bond)

Leadacetate test

Sulphosalicyclicacid test

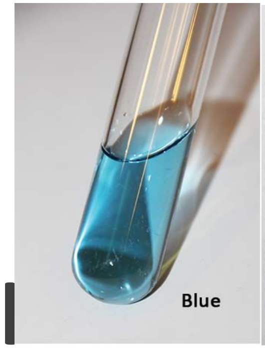


IDENTIFICATION OF INDIVIDUAL PROTEINS



BIURET TEST

(To detect presence of protein in gives samp



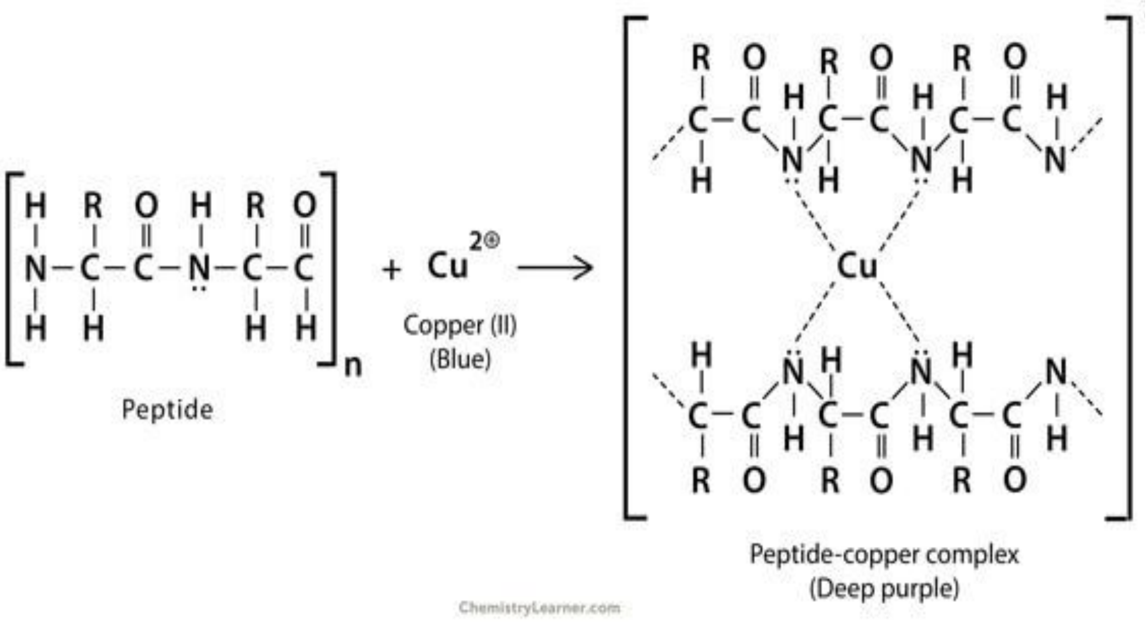
Negative biuret test

Positive biuret test

Principal: Alkaline $CuSO_4$ reacts with compounds containing two or more peptide bonds to give a violet colored product which is due to formation of co-ordination complex of cupric ions with unshared electron pairs of peptide nitrogen.

Procedure:

1. Take 1 ml of test solutions in dry test tube.
2. Add 1 ml of biuret reagent and mix well.
3. Observe the color of solution.
4. Deep blue/violet color indicates presence of proteins.

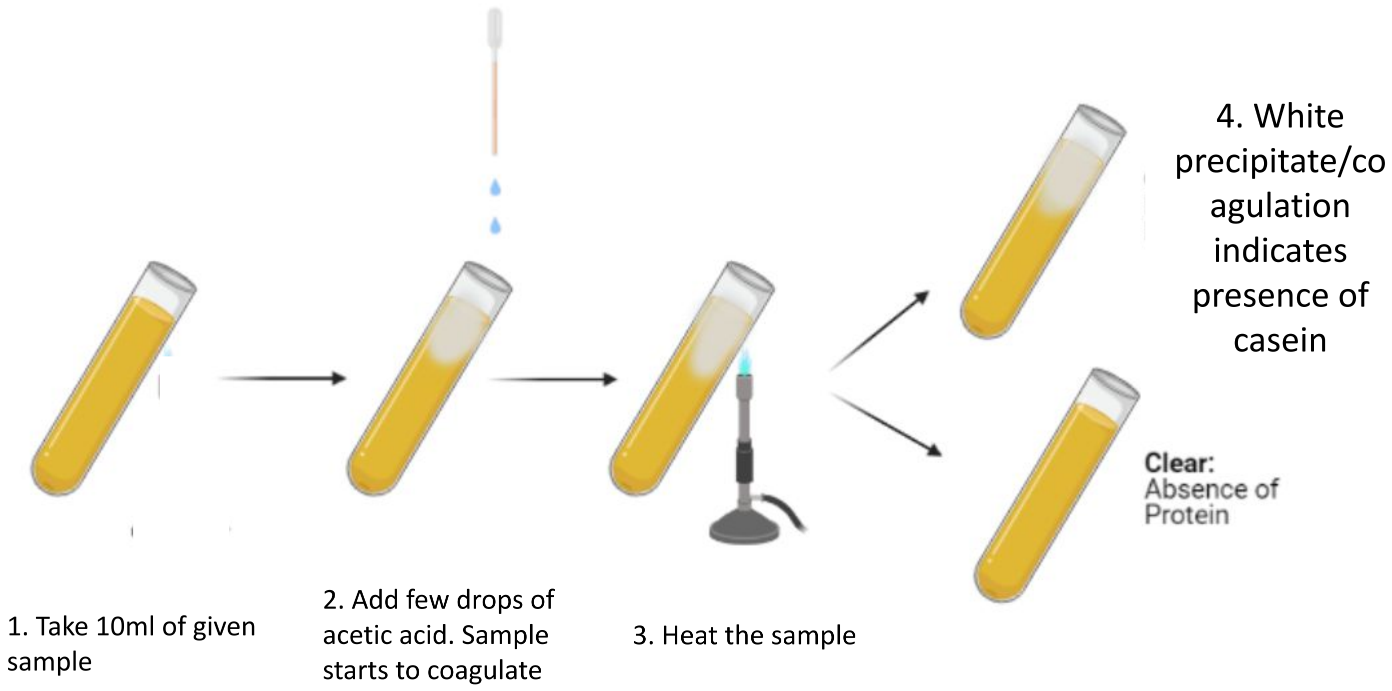


ACETIC ACID TEST

(To identify the presence of casein by acetic acid test)

Principal: Casein is a milk protein which has isoelectric point of pH 4.6. This test is based on the principle that casein get precipitated on addition of an acid (acidic medium).

Procedure:



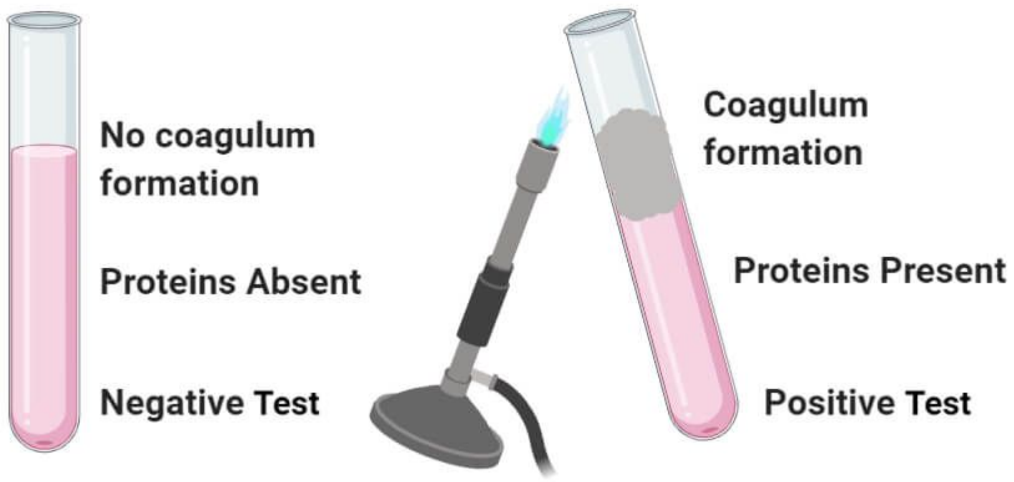
HEAT COAGULATION TEST

(To identify the presence of proteins)

Principal: Heat coagulation test is usually performed to detect presence of proteins in urine i.e. proteinuria. Upon heating protein structure denatures and coagulates, thus results in white cloudy precipitates. Test is positive for albumin and globulin.

Procedure:

1. Take 10 ml of test sample in a clean dry test tube.
2. Heat the sample for 4-5 min.
3. Observe the color/state of solution.
4. Coagulation of sample indicates presence of proteins.



THANK YOU